CSC 251 EXAM 1

STUDY GUIDE

TEST STRUCTURE

- You will have the entire 75 minute meeting time to complete the test.
- The test will include the following types of questions:
    - Short-answer
    - Arithmetic and Boolean logic
    - A short program, program fragments, or single lines of code to write or debug
    - Output of short coding blocks or programs
    - Fill-in-the-blank with vocabulary terms
    - Multiple choice and T/F
- Materials
    - You may bring with you one 3-inch × 5-inch note card with any notes you would like to have for the test written on ONE SIDE of the card. Note cards may not be shared between students, they must be handwritten, and they will be collected when you turn in your exam.
    - If you have a note card, your name should be clearly printed on it.
    - If your note card is too large, you will be required to trim it to 3'x5'. If you start taking the test with a note card that's too large, your score on the test will be penalized.
    - You may not use any textbook, website nor notes during the test.
    - You may not use any electronic devices such as cell phones or calculators during the test.
    - You will be expected to write your answers in the space below each question, on the FRONT of the test pages. The instructor attempts to provide adequate space for each question. No additional pages or scratch paper should be added to your test unless provided by the instructor. No answers written on the back of the page will be graded although you may use the back of the test pages for scratch work.
    - The first page of the test can be removed and used as a cover sheet while you complete your test.
- If you must leave the classroom for a bathroom break, you must notify and leave your cell phone with the instructor. Only one student at a time may leave the room at a time. Late in the test period, students may not be permitted to take a bathroom break.

EXAM 1 TOPICS

Any topics covered during lectures 01-09 may be asked on this exam. Questions might require you to use a particular programming construct, produce code using that construct, explain the construct, and make judgments about code using that type of construct.

- General understanding of the Python language, its uses, advantages, disadvantages, documentation
- Input statements
- Assignment statements, type conversion, type hints
- Python arithmetic, operator precedence
- Output statements, formatting output with specifiers and f-strings
- Conditional statements (if-elif-else, nested ifs), Boolean variables, logical expressions
- Iteration (while and for loops), range objects, input validation, augmented assignment, sentinels
- Functions (void and value-returning), global and local variables, passing arguments, return value(s), lambda functions
- Python sequences, comprehensions, 1D and 2D lists, conversion between sequences
- Operations on lists and strings - slicing, indexing, iterating, concatenating, searching, using with functions, methods for each

- Tuple operations – packing/unpacking, zipping
- Dictionaries - use, operations, and methods
- Sets – use, operations, methods
- Maps – use, creation, use of lambda functions
- Exception Handling – common exceptions, input validation
- File I/O – opening, reading from, writing to, and closing files; text and CVS files, processing

